Data communications (Lab 11): Frequency Domain Representation of a Sine Wave

Objective: To understand and simulate Go-Back-N Flow Control / ARQ (Automatic Repeat Request) and observe how data frames are transmitted, acknowledged, and retransmitted when an error occurs.

Theory

Go-Back-N is a sliding window flow control protocol used in data communication to ensure reliable transmission of frames between sender and receiver.

In this protocol:

- The sender can transmit **multiple frames continuously** without waiting for individual acknowledgements.
- A **window size (N)** determines how many frames can be sent before receiving ACKs.
- The receiver accepts frames **in order only**.
- If a frame is lost or damaged:
    - Receiver discards that frame and all subsequent frames.
    - Sender retransmits the lost frame **and all frames after it**.

This is why it is called **Go-Back-N**:
the sender goes back and retransmits from the erroneous frame onward.

Working Principle:

Example: Window size = 4; Sender sends: $F1, F2, F3, F4$; Suppose Frame 3 is lost: Receiver gets: $F1, F2, X, F4$
Receiver:

- accepts F1 and F2
- rejects F4 because F3 is missing
- sends ACK for last correctly received frame

Sender then retransmits: $F3, F4$

Algorithm:

1. Sender sends frames up to window size.
2. Receiver checks each frame.
3. If frame received correctly → ACK sent.
4. If frame lost/damaged → no ACK / negative ACK.
5. Sender timeout occurs.
6. Sender retransmits lost frame and all subsequent frames.

Matlab Code:

```
clc;
clear;

n = input('Enter total number of frames: ');
```

```
w = input('Enter window size: ');

i = 1;

while i <= n
    fprintf('\nSending frames: ');

    for j = i:min(i+w-1,n)
        fprintf('%d ', j);
    end

    fprintf('\nEnter frame lost (0 for none): ');
    lost = input('');

    if lost == 0
        fprintf('All frames acknowledged successfully.\n');
        i = i + w;
    else
        fprintf('Frame %d lost.\n', lost);
        fprintf('Go back and retransmit from Frame %d\n', lost);
        i = lost;
    end
end

fprintf('\nTransmission Completed Successfully.\n');
```

Code Explanation

1. Initialization

```
clc;
clear;
close all;
```

- `clc` → Clears command window
- `clear` → Removes all variables

2. Input Frames

```
n = input('Enter total number of frames: ');
w = input('Enter window size: ');
```

- `n` = total frames
- `w` = sender window size

3. Send Frames

```
for j = i:min(i+w-1,n)
```

Sends frames within window size.

Example: If: $i = 1; w = 4$ Frames sent: 1,2,3,4

4. Check Lost Frame

```
    lost = input('');
```

User enters lost frame number. Example: Frame 3 lost.

5. Retransmission

i = lost;

Transmission restarts from lost frame. Example: Resend: 3,4,5...

Result:

The MATLAB program simulates:

- ✓ frame transmission
- ✓ acknowledgement
- ✓ frame loss
- ✓ retransmission

It clearly demonstrates Go-Back-N operation.

Conclusion: The experiment successfully demonstrated Go-Back-N Flow Control. Frames were transmitted using a sliding window, and when a frame was lost, retransmission started from that frame onward, verifying the Go-Back-N protocol.